Staggered Roll-out of a New Recommendation Algorithm

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1 1. Visualization of Staggered Roll-Out and Watch Time

1.1 Data set description

```
df <- read.csv("streamflix.csv", stringsAsFactors = FALSE)
head(df, 5)
```

```
        country       week watch_time_per_user new_algo_deployed
1 UnitedKingdom 2021-01-04                4.82                 0
2 UnitedKingdom 2021-01-11                4.75                 0
3 UnitedKingdom 2021-01-18                5.10                 0
4 UnitedKingdom 2021-01-25                4.41                 0
5 UnitedKingdom 2021-02-01                5.18                 0
  marketing_spend avg_monthly_price broadband_penetration
1            1.43             11.67                   91.7
2            1.36             11.55                   91.4
3            1.37             11.54                   91.0
4            1.45             11.49                   91.5
5            1.40             11.68                   91.6
```

1.2 Visualization

```
library(panelView)
```

```
Warning: package 'panelView' was built under R version 4.5.1

## See bit.ly/panelview4r for more info.
## Report bugs -> yiqingxu@stanford.edu.
```

```
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag
```

```
The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
library(ggplot2)
library(lubridate)
```

```
Warning: package 'lubridate' was built under R version 4.5.1


Attaching package: 'lubridate'

The following objects are masked from 'package:base':

    date, intersect, setdiff, union
```

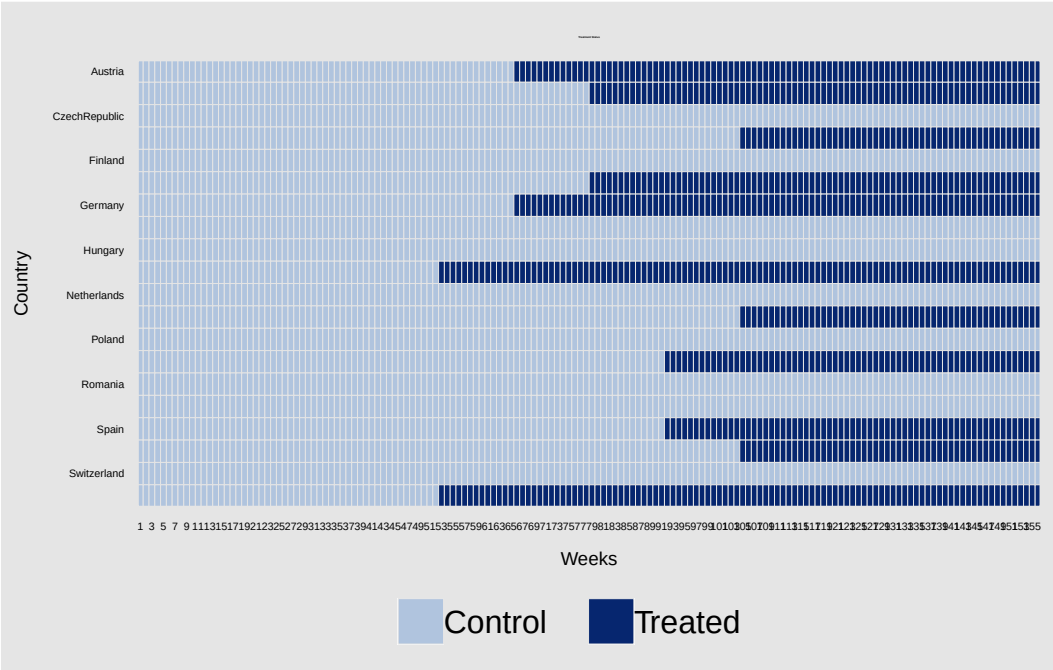
```r
# Converting 'week' column to Date objects
df$week <- ymd(df$week)

# Converting 'country' to a factor
df$country <- factor(df$country) # Corrected: Removed '$country' after df$country

# Creating a numeric 'time' variable for panelView
earliest_week <- min(df$week, na.rm = TRUE)
df$numeric_week <- as.numeric(difftime(df$week, earliest_week, units = "weeks"))+1
df$new_algo_deployed <- as.numeric(df$new_algo_deployed)


# Panel Plot using panelView
panelview(
  watch_time_per_user ~ new_algo_deployed,
  data = df,
  index = c("country", "numeric_week"),    # Specify unit and time IDs using 'index'
  xlab = "Weeks",
  ylab = "Country",
  cex.main = 1.2,
  cex.lab = 7,
  cex.axis = 4,
  legend.labs = c("Control", "Treated"),
  axis.lab.gap = c(1, 1)
)
```

```
Specified labels in the order of: Under Control, Under Treatment.
```



2 2. Two-Way Fixed Effects (TWFE) Estimation

2.1 Two-Way Fixed Effects model

```r
library(fixest)
```

```
Warning: package 'fixest' was built under R version 4.5.1
```

```r
mod_twfe <- feols(
  watch_time_per_user ~ new_algo_deployed + marketing_spend + avg_monthly_price + broadband_p
  cluster = ~ country + week,
  data = df )

summary(mod_twfe)
```

```
OLS estimation, Dep. Var.: watch_time_per_user
```

```
Observations: 3,120
Fixed-effects: country: 20,  week: 156
Standard-errors: Clustered (country & week)
                       Estimate Std. Error   t value  Pr(>|t|)
new_algo_deployed      0.468457   0.017517 26.743032 < 2.2e-16 ***
marketing_spend       -0.003101   0.053212 -0.058278   0.95414
avg_monthly_price      0.066630   0.125601  0.530485   0.60192
broadband_penetration  0.003357   0.024818  0.135253   0.89383
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 0.294896     Adj. R2: 0.745851
                 Within R2: 0.15799
```

2.2 Point estimate of the ATT and its SEs clustered by country & week.

```r
# Extracting coefficient (ATT) and clustered SE
att <- coef(mod_twfe)["new_algo_deployed"]
se <- sqrt(vcov(mod_twfe)["new_algo_deployed", "new_algo_deployed"])

# Report as formatted text
cat(sprintf("ATT : %.4f\n", att))
```

```
ATT : 0.4685
```

```r
cat(sprintf("Clustered SE: %.4f\n", se))
```

```
Clustered SE: 0.0175
```

```r
#at 95% confidence interval
lower <- att - 1.96 * se
upper <- att + 1.96 * se
cat(sprintf("95%% CI: [%.4f, %.4f]\n", lower, upper))
```

```
95% CI: [0.4341, 0.5028]
```

Weakness of TWFE:

The TWFE model assumes that all treated units are treated at the same time and that the treatment effect is the same across all units and time periods. In our setting, its staggered

rollout, meaning the treatment is introduced at different times across different countries. This violates the TWFE assumption that all treated units receive the treatment at the same time. As a result, TWFE may compare already treated countries to newly treated ones, leading to biased results.

3 3. Dynamic Two-Way Fixed Effects (Event Study) Estimation

3.1 Dynamic Two-Way Fixed Effects / Event Studies model including the covariates.

```
library(dplyr)

# Create a numeric time index
df <- df %>%
  mutate(time = as.integer(as.factor(week)))

# Get first treated week for each country
first_treat <- df %>%
  filter(new_algo_deployed == 1) %>%
  group_by(country) %>%
  summarise(treat_time = min(time))

# Merge treatment timing into main data
df <- df %>%
  left_join(first_treat, by = "country") %>%
  mutate(d_it_k = time - treat_time)

df$d_it_k[is.na(df$treat_time)] <- NA


#Event study model with covariates
model_eventstudy <- fixest::feols(watch_time_per_user ~ i(d_it_k, ref = c(-1, Inf)) +
                 marketing_spend + avg_monthly_price + broadband_penetration
               | country + time, data = df)
```

```
NOTE: 1,404 observations removed because of NA values (RHS: 1,404).

The variables 'd_it_k::91', 'd_it_k::92', 'd_it_k::93', 'd_it_k::94', 'd_it_k::95', 'd_it_k:
```

```r
summary(model_eventstudy)
```

```
OLS estimation, Dep. Var.: watch_time_per_user
Observations: 1,716
Fixed-effects: country: 11,  time: 156
Standard-errors: Clustered (country)
                      Estimate Std. Error   t value  Pr(>|t|)
d_it_k::-104          1.016897   0.231045  4.401288 0.0013324 **
d_it_k::-103          1.011972   0.383235  2.640608 0.0247082 *
d_it_k::-102          0.505658   0.600940  0.841444 0.4197656
d_it_k::-101          0.734998   0.340952  2.155720 0.0565163 .
d_it_k::-100          0.329489   0.423408  0.778182 0.4544797
d_it_k::-99           0.404154   0.619835  0.652035 0.5290798
d_it_k::-98          -0.162504   0.459645 -0.353542 0.7310222
d_it_k::-97           0.660032   0.398527  1.656177 0.1286835
d_it_k::-96           0.727360   0.348888  2.084796 0.0636831 .
d_it_k::-95           0.735199   0.262041  2.805659 0.0186130 *
d_it_k::-94           0.845503   0.476955  1.772711 0.1066822
d_it_k::-93           0.406341   0.473515  0.858137 0.4109130
d_it_k::-92           0.392840   0.300332  1.308018 0.2201354
d_it_k::-91           0.769470   0.343497  2.240107 0.0489954 *
d_it_k::-90           0.582849   0.335570  1.736891 0.1130507
d_it_k::-89           0.456668   0.542523  0.841748 0.4196035
d_it_k::-88           0.639839   0.444517  1.439404 0.1805947
d_it_k::-87           0.279573   0.369294  0.757047 0.4664845
d_it_k::-86           0.390734   0.536220  0.728682 0.4829119
d_it_k::-85          -0.290760   0.389892 -0.745744 0.4729876
d_it_k::-84           0.686258   0.281496  2.437897 0.0349746 *
d_it_k::-83           0.799144   0.366976  2.177644 0.0544619 .
d_it_k::-82           0.763763   0.257707  2.963686 0.0141983 *
d_it_k::-81           0.647539   0.411083  1.575202 0.1462872
d_it_k::-80           0.316654   0.393294  0.805134 0.4394660
d_it_k::-79           0.126121   0.332526  0.379281 0.7124084
d_it_k::-78           0.575739   0.255877  2.250065 0.0481744 *
d_it_k::-77           0.869496   0.347415  2.502756 0.0312987 *
d_it_k::-76           0.280349   0.514782  0.544597 0.5979643
d_it_k::-75           0.168981   0.392795  0.430202 0.6761722
d_it_k::-74           0.250259   0.420999  0.594442 0.5654221
d_it_k::-73           0.429595   0.536107  0.801323 0.4415686
d_it_k::-72           0.007092   0.438042  0.016191 0.9874008
d_it_k::-71           0.487888   0.281729  1.731759 0.1139910
d_it_k::-70           0.711768   0.341127  2.086521 0.0634990 .
```

d_it_k::-69	0.325903	0.263898	1.234958	0.2450755	
d_it_k::-68	0.709985	0.435537	1.630139	0.1341261	
d_it_k::-67	0.164056	0.443322	0.370062	0.7190537	
d_it_k::-66	0.284779	0.216577	1.314911	0.2178943	
d_it_k::-65	0.761636	0.196862	3.868878	0.0031149	**
d_it_k::-64	0.608304	0.306765	1.982966	0.0754993	.
d_it_k::-63	0.055915	0.461107	0.121263	0.9058848	
d_it_k::-62	0.146195	0.354915	0.411917	0.6890910	
d_it_k::-61	0.228236	0.400111	0.570433	0.5809743	
d_it_k::-60	0.412652	0.537428	0.767827	0.4603360	
d_it_k::-59	-0.172040	0.402206	-0.427741	0.6779050	
d_it_k::-58	0.430311	0.256159	1.679859	0.1239061	
d_it_k::-57	0.731334	0.332166	2.201709	0.0522896	.
d_it_k::-56	0.272993	0.221752	1.231076	0.2464625	
d_it_k::-55	0.719890	0.359407	2.002995	0.0730227	.
d_it_k::-54	0.197072	0.361114	0.545733	0.5972123	
d_it_k::-53	0.188078	0.186074	1.010772	0.3359570	
d_it_k::-52	0.626016	0.225589	2.775023	0.0196175	*
d_it_k::-51	0.558343	0.300644	1.857156	0.0929478	.
d_it_k::-50	0.207991	0.420388	0.494760	0.6314589	
d_it_k::-49	0.137336	0.376417	0.364852	0.7228200	
d_it_k::-48	0.216679	0.440117	0.492320	0.6331217	
d_it_k::-47	0.368017	0.512457	0.718142	0.4891080	
d_it_k::-46	-0.144441	0.362045	-0.398959	0.6983103	
d_it_k::-45	0.312203	0.227055	1.375009	0.1991501	
d_it_k::-44	0.613584	0.334059	1.836755	0.0961079	.
d_it_k::-43	0.252732	0.263146	0.960426	0.3594873	
d_it_k::-42	0.797313	0.365039	2.184184	0.0538632	.
d_it_k::-41	-0.091671	0.372860	-0.245858	0.8107650	
d_it_k::-40	0.090847	0.233637	0.388838	0.7055467	
d_it_k::-39	0.556265	0.183149	3.037230	0.0125216	*
d_it_k::-38	0.463976	0.222945	2.081122	0.0640770	.
d_it_k::-37	0.060815	0.379850	0.160104	0.8759871	
d_it_k::-36	0.330053	0.295048	1.118640	0.2894449	
d_it_k::-35	0.160008	0.410321	0.389958	0.7047442	
d_it_k::-34	0.292444	0.538092	0.543483	0.5987031	
d_it_k::-33	-0.362684	0.392644	-0.923698	0.3773935	
d_it_k::-32	0.200825	0.324701	0.618491	0.5500790	
d_it_k::-31	0.497204	0.244429	2.034141	0.0693231	.
d_it_k::-30	0.160093	0.192294	0.832540	0.4245406	
d_it_k::-29	0.631932	0.338122	1.868948	0.0911654	.
d_it_k::-28	-0.046109	0.398494	-0.115708	0.9101748	
d_it_k::-27	0.156188	0.165378	0.944432	0.3672080	

d_it_k::-26	0.434410	0.204725	2.121913	0.0598305	.
d_it_k::-25	0.176531	0.348822	0.506078	0.6237711	
d_it_k::-24	0.084470	0.353779	0.238766	0.8161092	
d_it_k::-23	0.110164	0.303884	0.362520	0.7245083	
d_it_k::-22	-0.034300	0.293372	-0.116916	0.9092415	
d_it_k::-21	0.478770	0.470647	1.017260	0.3330098	
d_it_k::-20	-0.485000	0.273094	-1.775942	0.1061241	
d_it_k::-19	0.210696	0.252259	0.835236	0.4230912	
d_it_k::-18	0.340319	0.307404	1.107072	0.2941818	
d_it_k::-17	0.154835	0.171591	0.902349	0.3880894	
d_it_k::-16	0.441172	0.274773	1.605590	0.1394447	
d_it_k::-15	-0.118862	0.324222	-0.366606	0.7215506	
d_it_k::-14	0.137614	0.098001	1.404206	0.1905457	
d_it_k::-13	0.340105	0.202517	1.679391	0.1239989	
d_it_k::-12	0.115747	0.229655	0.504003	0.6251772	
d_it_k::-11	0.036331	0.292749	0.124104	0.9036920	
d_it_k::-10	0.132729	0.233055	0.569517	0.5815721	
d_it_k::-9	0.082627	0.240707	0.343266	0.7385053	
d_it_k::-8	0.099000	0.395392	0.250385	0.8073585	
d_it_k::-7	-0.394362	0.323128	-1.220453	0.2502907	
d_it_k::-6	0.193427	0.243436	0.794569	0.4453117	
d_it_k::-5	0.062323	0.328264	0.189857	0.8532196	
d_it_k::-4	0.045368	0.131632	0.344660	0.7374889	
d_it_k::-3	0.691688	0.318020	2.174980	0.0547077	.
d_it_k::-2	-0.279017	0.356563	-0.782518	0.4520423	
d_it_k::0	0.557618	0.154588	3.607128	0.0047909	**
d_it_k::1	0.866502	0.216689	3.998833	0.0025231	**
d_it_k::2	0.436951	0.364227	1.199665	0.2579208	
d_it_k::3	0.577569	0.164730	3.506160	0.0056679	**
d_it_k::4	0.533935	0.211599	2.523334	0.0302142	*
d_it_k::5	0.717150	0.319118	2.247285	0.0484022	*
d_it_k::6	-0.059060	0.184321	-0.320420	0.7552451	
d_it_k::7	0.367100	0.291023	1.261413	0.2357907	
d_it_k::8	0.612549	0.329995	1.856240	0.0930876	.
d_it_k::9	0.523621	0.220560	2.374056	0.0390063	*
d_it_k::10	0.965206	0.311323	3.100342	0.0112441	*
d_it_k::11	0.041252	0.330135	0.124954	0.9030363	
d_it_k::12	0.345522	0.173660	1.989647	0.0746644	.
d_it_k::13	0.653662	0.138077	4.734024	0.0007993	***
d_it_k::14	0.683734	0.198976	3.436272	0.0063711	**
d_it_k::15	0.190640	0.301799	0.631676	0.5417683	
d_it_k::16	0.495589	0.156456	3.167596	0.0100285	*
d_it_k::17	0.441581	0.206070	2.142869	0.0577550	.

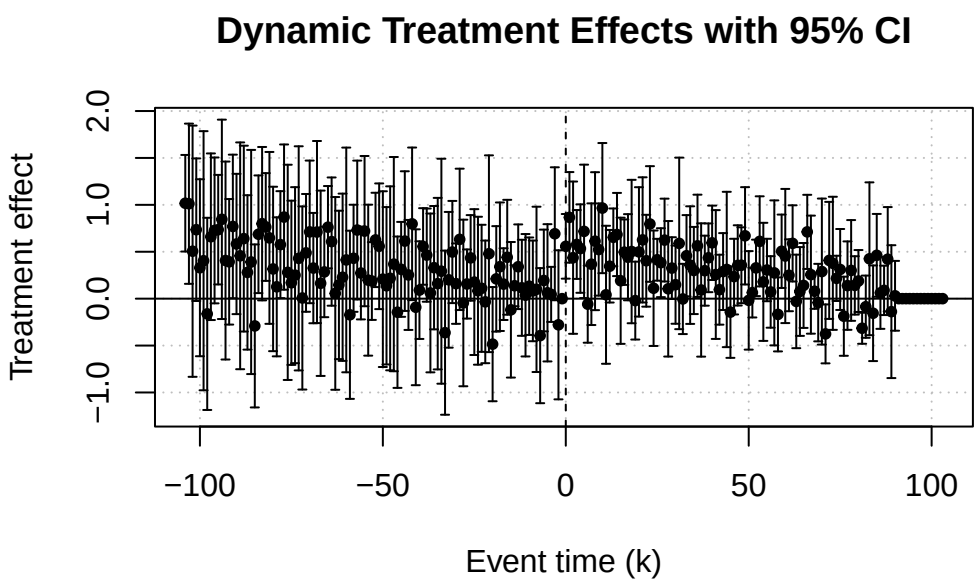
d_it_k::18	0.502156	0.343065	1.463734	0.1739795	
d_it_k::19	-0.020787	0.186324	-0.111562	0.9133786	
d_it_k::20	0.498355	0.309490	1.610245	0.1384220	
d_it_k::21	0.626397	0.299419	2.092043	0.0629128	.
d_it_k::22	0.404109	0.219639	1.839882	0.0956170	.
d_it_k::23	0.795880	0.276588	2.877493	0.0164561	*
d_it_k::24	0.110977	0.276112	0.401928	0.6961932	
d_it_k::25	0.414777	0.100016	4.147100	0.0019892	**
d_it_k::26	0.383388	0.164866	2.325458	0.0423769	*
d_it_k::27	0.624220	0.198169	3.149935	0.0103341	*
d_it_k::28	0.105145	0.324144	0.324377	0.7523359	
d_it_k::29	0.325554	0.147689	2.204320	0.0520590	.
d_it_k::30	0.148858	0.186885	0.796521	0.4442277	
d_it_k::31	0.589042	0.410290	1.435674	0.1816278	
d_it_k::32	-0.003081	0.168021	-0.018337	0.9857308	
d_it_k::33	0.457463	0.194309	2.354306	0.0403433	*
d_it_k::34	0.357504	0.281279	1.270996	0.2324991	
d_it_k::35	0.300037	0.208436	1.439466	0.1805776	
d_it_k::36	0.563140	0.244581	2.302468	0.0440688	*
d_it_k::37	0.092410	0.318918	0.289763	0.7779168	
d_it_k::38	0.298231	0.167453	1.780990	0.1052576	
d_it_k::39	0.435388	0.164399	2.648356	0.0243817	*
d_it_k::40	0.595276	0.177930	3.345569	0.0074205	**
d_it_k::41	0.257083	0.308082	0.834463	0.4235062	
d_it_k::42	0.095604	0.167304	0.571440	0.5803170	
d_it_k::43	0.288668	0.183985	1.568980	0.1477244	
d_it_k::44	0.311066	0.371535	0.837243	0.4220138	
d_it_k::45	-0.143744	0.218694	-0.657283	0.5258376	
d_it_k::46	0.234907	0.180963	1.298094	0.2233949	
d_it_k::47	0.354324	0.192039	1.845064	0.0948089	.
d_it_k::48	0.356044	0.181533	1.961322	0.0782644	.
d_it_k::49	0.669872	0.232774	2.877782	0.0164479	*
d_it_k::50	-0.018182	0.235645	-0.077159	0.9400192	
d_it_k::51	0.064400	0.118903	0.541619	0.5999397	
d_it_k::52	0.327992	0.134968	2.430151	0.0354410	*
d_it_k::53	0.610608	0.215571	2.832511	0.0177752	*
d_it_k::54	0.178938	0.282452	0.633515	0.5406155	
d_it_k::55	0.305317	0.149302	2.044965	0.0680796	.
d_it_k::56	0.071512	0.171058	0.418055	0.6847424	
d_it_k::57	0.274657	0.346787	0.792004	0.4467389	
d_it_k::58	-0.167881	0.177080	-0.948055	0.3654486	
d_it_k::59	0.505144	0.175530	2.877824	0.0164467	*
d_it_k::60	0.454523	0.320945	1.416201	0.1871032	

```
d_it_k::61             0.249297   0.171992  1.449469 0.1778323
d_it_k::62             0.589067   0.182288  3.231511 0.0089979 **
d_it_k::63            -0.029522   0.223734 -0.131951 0.8976402
d_it_k::64             0.076319   0.212425  0.359275 0.7268601
d_it_k::65             0.140566   0.203062  0.692232 0.5045476
d_it_k::66             0.711805   0.177331  4.014003 0.0024621 **
d_it_k::67             0.255850   0.282421  0.905917 0.3862871
d_it_k::68             0.078966   0.084089  0.939078 0.3698189
d_it_k::69            -0.042170   0.185331 -0.227539 0.8245896
d_it_k::70             0.288682   0.349305  0.826446 0.4278298
d_it_k::71            -0.375668   0.140962 -2.665038 0.0236935 *
d_it_k::72             0.407843   0.279951  1.456835 0.1758338
d_it_k::73             0.376643   0.318404  1.182911 0.2642045
d_it_k::74             0.216177   0.108156  1.998755 0.0735404 .
d_it_k::75             0.310911   0.193696  1.605151 0.1395416
d_it_k::76            -0.187306   0.189210 -0.989939 0.3455516
d_it_k::77             0.138957   0.211532  0.656907 0.5260692
d_it_k::78             0.300752   0.241009  1.247885 0.2405023
d_it_k::79             0.141125   0.139657  1.010508 0.3360773
d_it_k::80             0.188093   0.148136  1.269733 0.2329307
d_it_k::81            -0.315618   0.074243 -4.251143 0.0016865 **
d_it_k::82            -0.085850   0.148809 -0.576913 0.5767536
d_it_k::83             0.424985   0.365570  1.162530 0.2720128
d_it_k::84            -0.155684   0.228797 -0.680448 0.5116669
d_it_k::85             0.461691   0.198075  2.330893 0.0419863 *
d_it_k::86             0.061599   0.172090  0.357946 0.7278240
d_it_k::87             0.088227   0.118087  0.747136 0.4721832
d_it_k::88             0.420277   0.249455  1.684779 0.1229338
d_it_k::89            -0.138620   0.317798 -0.436191 0.6719646
d_it_k::90             0.030437   0.166991  0.182266 0.8590156
marketing_spend        0.024758   0.091937  0.269289 0.7931821
avg_monthly_price     -0.077290   0.214841 -0.359755 0.7265115
broadband_penetration  0.035765   0.039870  0.897038 0.3907831
... 13 variables were removed because of collinearity (d_it_k::91, d_it_k::92 and 11 others
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 0.279582     Adj. R2: 0.742815
                 Within R2: 0.217972
```

3.2 Dynamic treatment effects with 95% confidence bands

```
library(fixest)

iplot(
  model_eventstudy,
  ref.line = 0,
  xlab = "Event time (k)",           # Label for x-axis
  ylab = "Treatment effect",         # Label for y-axis
  main = "Dynamic Treatment Effects with 95% CI"  # Plot title
)
```



<u>Weakness of dynamic TWFE (Event study):</u>

The main limitation of the dynamic TWFE (event study) model in a staggered rollout setting is that it assumes the treatment has the same effect for all countries and at all times. In our case, the algorithm was introduced at different points in time across countries, but the model ignores this variation. It treats early-treated and late-treated units as if they are directly comparable, which can lead to biased result.

4 4. Callaway & Sant'Anna (2021) Estimation

4.1 Callaway & Sant'Anna (2021) estimator for staggered treatment timing including the covariates.

```
library(did)
```

```
Warning: package 'did' was built under R version 4.5.1
```

```
library(dplyr)
library(lubridate)
library(ggplot2)
# Convert 'week' column to Date objects
df$week <- ymd(df$week)

df$country_id_numeric <- as.numeric(factor(df$country))

# This portion of code is done with the help of some LLM to Create a numeric 'time' variable
earliest_week <- min(df$week, na.rm = TRUE)
df$numeric_week <- as.numeric(difftime(df$week, earliest_week, units = "weeks")) + 1

df_processed <- df %>%
  group_by(country_id_numeric) %>%
  mutate(
    group = ifelse(
      any(new_algo_deployed == 1),
      min(numeric_week[new_algo_deployed == 1]),
      Inf
    )
  ) %>%
  ungroup()
# Ensure 'new_algo_deployed' is numeric (0 or 1)
df_processed$new_algo_deployed <- as.numeric(df_processed$new_algo_deployed)



#Callaway & Sant'Anna (2021) Model

drdid_results <- att_gt(
  yname = "watch_time_per_user",
```

```
  tname = "numeric_week",
  idname = "country_id_numeric", # Use the new numeric ID column
  gname = "group",
  xformla = ~ 1, # No covariates
  data = df_processed,
  allow_unbalanced_panel = TRUE,
  control_group = "nevertreated",
  anticipation = 0,
  clustervars = "country_id_numeric",
  est_method = "dr"
)
```

```
Warning in pre_process_did(yname = yname, tname = tname, idname = idname, : Be aware that the
  Check groups: 53,66,79,92,105.

Warning in att_gt(yname = "watch_time_per_user", tname = "numeric_week", : Not
returning pre-test Wald statistic due to singular covariance matrix
```

```
cat("\n Group-Time ATTs (ATT_gt) \n")
```

```
 Group-Time ATTs (ATT_gt)
```

```
print(drdid_results$att.egt)
```

```
NULL
```

```
# Aggregate to get overall ATT
overall_att <- aggte(drdid_results, type = "group")
cat("\ nOverall ATT (Aggregated) \n")
```

```
 nOverall ATT (Aggregated)
```

```
print(overall_att)
```

```
Call:
aggte(MP = drdid_results, type = "group")

Reference: Callaway, Brantly and Pedro H.C. Sant'Anna.  "Difference-in-Differences with Multi


Overall summary of ATT's based on group/cohort aggregation:
    ATT    Std. Error     [ 95%  Conf. Int.]
 0.4327        0.1005     0.2357      0.6297 *


Group Effects:
 Group Estimate Std. Error [95% Simult.  Conf. Band]
    53   0.3900     0.0752        0.2240      0.5561 *
    66   0.3712     0.1553        0.0282      0.7141 *
    79   0.6477     0.2778        0.0341      1.2613 *
    92   0.4626     0.1492        0.1329      0.7923 *
   105   0.3388     0.1595       -0.0135      0.6910
---
Signif. codes: `*' confidence band does not cover 0

Control Group:  Never Treated,  Anticipation Periods:  0
Estimation Method:  Doubly Robust
```

```r
# Aggregate to get dynamic effects (ATTs over time)
dynamic_att <- aggte(drdid_results, type = "dynamic")
cat("\n Dynamic ATTs (Aggregated over time) \n")
```

```
 Dynamic ATTs (Aggregated over time)
```

```r
print(dynamic_att)
```

```
Call:
aggte(MP = drdid_results, type = "dynamic")

Reference: Callaway, Brantly and Pedro H.C. Sant'Anna.  "Difference-in-Differences with Multi
```

```
Overall summary of ATT's based on event-study/dynamic aggregation:
    ATT    Std. Error     [ 95%  Conf. Int.]
 0.4256        0.1023     0.2251       0.626 *


Dynamic Effects:
 Event time Estimate Std. Error [95% Simult.  Conf. Band]
       -103  -0.2289     0.2184       -0.8770      0.4193
       -102   0.0856     0.2725       -0.7234      0.8945
       -101   0.2044     0.2160       -0.4366      0.8455
       -100  -0.2122     0.2002       -0.8066      0.3821
        -99  -0.0433     0.2076       -0.6594      0.5728
        -98  -0.0700     0.2356       -0.7692      0.6292
        -97   0.2189     0.2504       -0.5243      0.9621
        -96   0.0067     0.2680       -0.7887      0.8020
        -95   0.1444     0.3340       -0.8471      1.1359
        -94  -0.4367     0.1296       -0.8213     -0.0520 *
        -93   0.3144     0.1845       -0.2332      0.8621
        -92  -0.0433     0.2982       -0.9284      0.8417
        -91  -0.0044     0.2416       -0.7216      0.7127
        -90  -0.2822     0.2094       -0.9039      0.3394
        -89   0.2982     0.1224       -0.0651      0.6615
        -88   0.2451     0.2157       -0.3951      0.8853
        -87  -0.2249     0.1893       -0.7866      0.3369
        -86  -0.1013     0.1718       -0.6114      0.4087
        -85  -0.0833     0.1652       -0.5736      0.4069
        -84   0.3242     0.2098       -0.2985      0.9469
        -83  -0.0640     0.1953       -0.6437      0.5157
        -82   0.1324     0.1352       -0.2688      0.5337
        -81  -0.5360     0.1059       -0.8503     -0.2217 *
        -80   0.4018     0.1535       -0.0539      0.8574
        -79  -0.1487     0.3113       -1.0727      0.7753
        -78  -0.1324     0.2123       -0.7624      0.4976
        -77   0.2221     0.2115       -0.4056      0.8497
        -76  -0.0760     0.1486       -0.5171      0.3650
        -75  -0.1386     0.2137       -0.7729      0.4958
        -74   0.2362     0.1764       -0.2873      0.7597
        -73  -0.1459     0.1113       -0.4763      0.1845
        -72   0.2248     0.1899       -0.3388      0.7884
        -71  -0.1605     0.1671       -0.6564      0.3354
        -70   0.0868     0.1596       -0.3869      0.5606
        -69  -0.1521     0.1914       -0.7203      0.4162
        -68  -0.1597     0.3229       -1.1181      0.7987
```

-67	0.2330	0.1861	-0.3193	0.7853
-66	-0.0244	0.1582	-0.4940	0.4451
-65	0.2529	0.1589	-0.2186	0.7244
-64	-0.1370	0.1532	-0.5918	0.3177
-63	-0.1832	0.2170	-0.8272	0.4608
-62	0.0838	0.1802	-0.4511	0.6187
-61	0.1384	0.1665	-0.3558	0.6325
-60	-0.0549	0.1376	-0.4632	0.3534
-59	0.0260	0.1476	-0.4121	0.4642
-58	-0.0120	0.1619	-0.4924	0.4685
-57	0.1825	0.1059	-0.1319	0.4969
-56	-0.2107	0.1769	-0.7359	0.3144
-55	-0.0384	0.1240	-0.4066	0.3298
-54	0.1658	0.1156	-0.1774	0.5090
-53	-0.1289	0.1345	-0.5282	0.2704
-52	0.1280	0.1894	-0.4340	0.6901
-51	-0.0483	0.1033	-0.3550	0.2585
-50	0.0360	0.1218	-0.3256	0.3975
-49	-0.1359	0.1352	-0.5373	0.2656
-48	0.1972	0.1356	-0.2052	0.5995
-47	-0.1267	0.1328	-0.5209	0.2676
-46	0.1356	0.1402	-0.2806	0.5517
-45	-0.1594	0.1159	-0.5035	0.1847
-44	0.1399	0.1185	-0.2117	0.4915
-43	-0.1190	0.1089	-0.4422	0.2042
-42	0.0997	0.1137	-0.2379	0.4373
-41	-0.1849	0.1461	-0.6187	0.2488
-40	0.0677	0.1746	-0.4505	0.5858
-39	0.1418	0.1611	-0.3363	0.6199
-38	-0.0514	0.1169	-0.3983	0.2955
-37	-0.0692	0.1479	-0.5081	0.3697
-36	0.2264	0.1363	-0.1782	0.6309
-35	-0.1039	0.1582	-0.5735	0.3656
-34	-0.1651	0.1363	-0.5696	0.2395
-33	0.0494	0.1563	-0.4146	0.5134
-32	-0.0579	0.2188	-0.7073	0.5916
-31	0.1533	0.1916	-0.4155	0.7222
-30	-0.0531	0.1535	-0.5086	0.4024
-29	-0.0329	0.1793	-0.5653	0.4994
-28	0.0451	0.1745	-0.4728	0.5629
-27	0.0583	0.1421	-0.3636	0.4802
-26	0.0271	0.1242	-0.3415	0.3957
-25	-0.2847	0.1575	-0.7522	0.1827

-24	0.2715	0.1251	-0.0999	0.6429	
-23	-0.0865	0.1660	-0.5793	0.4063	
-22	-0.0518	0.1672	-0.5482	0.4446	
-21	0.2019	0.1368	-0.2040	0.6079	
-20	-0.3089	0.0962	-0.5944	-0.0234	*
-19	0.1590	0.1541	-0.2985	0.6165	
-18	0.0647	0.1153	-0.2774	0.4069	
-17	-0.0242	0.1293	-0.4080	0.3595	
-16	-0.1972	0.1697	-0.7008	0.3065	
-15	0.1582	0.1647	-0.3306	0.6470	
-14	0.0462	0.1608	-0.4311	0.5235	
-13	-0.0143	0.1489	-0.4562	0.4275	
-12	-0.3001	0.1583	-0.7700	0.1698	
-11	0.3795	0.1379	-0.0299	0.7889	
-10	-0.0519	0.1142	-0.3907	0.2869	
-9	0.0191	0.1331	-0.3761	0.4143	
-8	-0.2077	0.1575	-0.6752	0.2599	
-7	0.0361	0.2394	-0.6745	0.7466	
-6	0.1059	0.1407	-0.3119	0.5236	
-5	-0.1953	0.1357	-0.5980	0.2075	
-4	0.1866	0.1558	-0.2760	0.6491	
-3	0.1598	0.1460	-0.2736	0.5932	
-2	-0.2502	0.1049	-0.5617	0.0613	
-1	0.0954	0.1524	-0.3569	0.5476	
0	0.3665	0.1566	-0.0983	0.8312	
1	0.6244	0.1679	0.1262	1.1227	*
2	0.5490	0.1406	0.1315	0.9664	*
3	0.4977	0.1350	0.0970	0.8984	*
4	0.5849	0.1448	0.1551	1.0148	*
5	0.4931	0.1584	0.0230	0.9633	*
6	0.3256	0.1291	-0.0577	0.7088	
7	0.2700	0.2002	-0.3243	0.8643	
8	0.4272	0.1448	-0.0025	0.8568	
9	0.5112	0.1219	0.1495	0.8729	*
10	0.5159	0.1402	0.0998	0.9319	*
11	0.3513	0.1531	-0.1030	0.8056	
12	0.4424	0.1724	-0.0692	0.9540	
13	0.5343	0.1647	0.0454	1.0233	*
14	0.4504	0.1370	0.0438	0.8570	*
15	0.3849	0.1622	-0.0965	0.8664	
16	0.5348	0.1779	0.0068	1.0629	*
17	0.5910	0.1250	0.2199	0.9621	*
18	0.3282	0.1615	-0.1513	0.8076	

19	0.4213	0.1765	-0.1025	0.9451	
20	0.4898	0.1231	0.1244	0.8552	*
21	0.5122	0.1307	0.1244	0.9001	*
22	0.4760	0.1486	0.0349	0.9171	*
23	0.4288	0.1671	-0.0672	0.9248	
24	0.4410	0.1526	-0.0120	0.8940	
25	0.5515	0.1009	0.2519	0.8511	*
26	0.3468	0.1361	-0.0572	0.7508	
27	0.4227	0.2007	-0.1729	1.0183	
28	0.3727	0.1191	0.0191	0.7263	*
29	0.4027	0.1102	0.0755	0.7299	*
30	0.3295	0.1662	-0.1637	0.8227	
31	0.4482	0.1337	0.0514	0.8449	*
32	0.3943	0.1563	-0.0696	0.8583	
33	0.3768	0.1720	-0.1338	0.8874	
34	0.3385	0.1239	-0.0293	0.7063	
35	0.4035	0.1396	-0.0108	0.8179	
36	0.3089	0.1536	-0.1471	0.7649	
37	0.4405	0.1746	-0.0777	0.9588	
38	0.5101	0.1504	0.0636	0.9566	*
39	0.4473	0.1416	0.0268	0.8677	*
40	0.3747	0.1323	-0.0179	0.7674	
41	0.5025	0.1460	0.0690	0.9360	*
42	0.3244	0.1921	-0.2458	0.8947	
43	0.5120	0.1411	0.0934	0.9307	*
44	0.2873	0.1189	-0.0657	0.6402	
45	0.3061	0.1260	-0.0680	0.6801	
46	0.2323	0.2274	-0.4427	0.9073	
47	0.3141	0.1421	-0.1078	0.7360	
48	0.5706	0.1191	0.2172	0.9240	*
49	0.5242	0.1609	0.0468	1.0017	*
50	0.4658	0.1312	0.0763	0.8552	*
51	0.3025	0.1322	-0.0898	0.6948	
52	0.4789	0.1136	0.1416	0.8162	*
53	0.5669	0.1386	0.1557	0.9782	*
54	0.5108	0.0932	0.2343	0.7874	*
55	0.6136	0.1003	0.3160	0.9112	*
56	0.3922	0.2097	-0.2303	1.0147	
57	0.2931	0.1784	-0.2366	0.8227	
58	0.4246	0.1719	-0.0858	0.9350	
59	0.5971	0.2004	0.0022	1.1920	*
60	0.5176	0.1712	0.0095	1.0258	*
61	0.5399	0.1983	-0.0486	1.1284	

62	0.5801	0.1702	0.0750	1.0853	*
63	0.5314	0.1570	0.0653	0.9975	*
64	0.4385	0.1779	-0.0896	0.9665	
65	0.3344	0.2634	-0.4473	1.1162	
66	0.6107	0.1715	0.1017	1.1198	*
67	0.6502	0.2103	0.0259	1.2745	*
68	0.5322	0.1940	-0.0437	1.1081	
69	0.3707	0.1861	-0.1818	0.9233	
70	0.3974	0.2611	-0.3775	1.1723	
71	0.2494	0.1954	-0.3307	0.8295	
72	0.5335	0.2613	-0.2421	1.3092	
73	0.5217	0.1767	-0.0027	1.0461	
74	0.6107	0.2220	-0.0481	1.2696	
75	0.3915	0.2744	-0.4231	1.2061	
76	0.4059	0.2203	-0.2479	1.0598	
77	0.5409	0.2564	-0.2202	1.3021	
78	0.4831	0.3587	-0.5817	1.5478	
79	-0.0394	0.1840	-0.5856	0.5067	
80	0.4894	0.3602	-0.5798	1.5587	
81	0.1272	0.2741	-0.6864	0.9409	
82	0.2414	0.1374	-0.1666	0.6494	
83	0.5633	0.1639	0.0768	1.0498	*
84	0.3533	0.1078	0.0335	0.6732	*
85	0.4492	0.1553	-0.0117	0.9100	
86	0.2242	0.2376	-0.4812	0.9295	
87	0.4789	0.2179	-0.1678	1.1255	
88	0.4867	0.3032	-0.4134	1.3868	
89	0.4617	0.3031	-0.4380	1.3614	
90	0.3861	0.2885	-0.4703	1.2425	
91	0.2678	0.1444	-0.1609	0.6964	
92	-0.2239	0.1878	-0.7812	0.3334	
93	0.2272	0.1955	-0.3531	0.8076	
94	0.6850	0.0717	0.4723	0.8977	*
95	0.4328	0.1649	-0.0567	0.9223	
96	0.3600	0.3394	-0.6473	1.3673	
97	0.6311	0.1237	0.2638	0.9984	*
98	0.2861	0.1230	-0.0788	0.6511	
99	0.1183	0.2422	-0.6004	0.8371	
100	0.4950	0.1507	0.0476	0.9424	*
101	0.1139	0.0920	-0.1592	0.3870	
102	0.8500	0.1623	0.3684	1.3316	*
103	0.3167	0.1565	-0.1478	0.7812	

```
Signif. codes: `*' confidence band does not cover 0

Control Group:  Never Treated,  Anticipation Periods:  0
Estimation Method:  Doubly Robust
```

5 5. Augmented Synthetic Control for Staggered Adoption

5.1 Augmented Synthetic Control Method

```r
library(augsynth)
library(dplyr)

# Panel prep (use exactly this block)
panel <- df %>%
  mutate(
    country  = as.factor(country),
    time_idx = as.integer(as.factor(week))
  )

# Augmented Synthetic Control on your data
asyn <- augsynth(
  watch_time_per_user ~ new_algo_deployed | marketing_spend + avg_monthly_price + broadband_p
  unit     = country,
  time     = time_idx,
  data     = panel,
  progfunc = "Ridge",
  scm      = TRUE,
  fixedeff = TRUE,   # allow intercept shifts
  alpha    = 0.5  #(partial pooling method)
)
```

```
More than one treatment time found. Running multisynth.

Warning in augsynth(watch_time_per_user ~ new_algo_deployed | marketing_spend +
: `progfunc` is not an argument for multisynth, so it is ignored
```

```r
# Results
summary(asyn)
```

```
Call:
multisynth(form = form, unit = !!enquo(unit), time = !!enquo(time),
    data = data, fixedeff = TRUE, scm = TRUE, progfunc = "Ridge",
    alpha = 0.5)

Average ATT Estimate (Std. Error): 0.460  (0.132)

Global L2 Imbalance: 0.082
Scaled Global L2 Imbalance: 0.210
Percent improvement from uniform global weights: 79

Individual L2 Imbalance: 0.325
Scaled Individual L2 Imbalance: 0.450
Percent improvement from uniform individual weights: 55

 Time Since Treatment   Level  Estimate Std.Error lower_bound upper_bound
                    0 Average 0.3901059 0.1474214  0.11845012   0.6934220
                    1 Average 0.6177279 0.2497353  0.20166993   1.1609954
                    2 Average 0.5362803 0.1974949  0.19838560   0.9491902
                    3 Average 0.5744626 0.1681975  0.27287273   0.9031790
                    4 Average 0.6405305 0.2061907  0.27163965   1.0477785
                    5 Average 0.4998882 0.1899486  0.16158726   0.8807557
                    6 Average 0.3279025 0.1489648  0.06811827   0.6560808
                    7 Average 0.2948740 0.1633721 -0.02775685   0.6198810
                    8 Average 0.4231504 0.1636940  0.12660491   0.7432085
                    9 Average 0.5590363 0.1701386  0.24681451   0.9315256
                   10 Average 0.5964327 0.1951270  0.22755481   0.9893237
                   11 Average 0.4192214 0.1629881  0.09686900   0.7269242
                   12 Average 0.4295612 0.1873785  0.08451618   0.8143212
                   13 Average 0.5904798 0.1775331  0.26897065   0.9469232
                   14 Average 0.4122850 0.1880839  0.07530183   0.8029098
                   15 Average 0.3645546 0.1619944  0.07225427   0.6872321
                   16 Average 0.5204411 0.2085556  0.15150904   0.9470354
                   17 Average 0.6539533 0.1968321  0.28916543   1.0465128
                   18 Average 0.3305445 0.1433976  0.06228847   0.6242747
                   19 Average 0.3513645 0.1919809 -0.02781695   0.7288778
                   20 Average 0.5679527 0.1834598  0.22706956   0.9252403
                   21 Average 0.5807362 0.1728130  0.28106931   0.9408117
                   22 Average 0.4328286 0.1821548  0.09010277   0.7919983
                   23 Average 0.4492034 0.1719995  0.11062478   0.7889040
                   24 Average 0.5693667 0.1747552  0.24370461   0.9139632
                   25 Average 0.5996083 0.1959024  0.24974523   0.9856988
```

```
26 Average 0.3728416 0.1629608  0.06668041   0.6923812
27 Average 0.4461089 0.1925270  0.08488286   0.8198260
28 Average 0.3693952 0.1520410  0.10164493   0.6958658
29 Average 0.4060388 0.1817664  0.07876984   0.7820968
30 Average 0.4077231 0.1577462  0.10343577   0.7352192
31 Average 0.4861474 0.1727965  0.17054027   0.8488555
32 Average 0.3651680 0.1820348  0.03371140   0.7307597
33 Average 0.4674085 0.1360977  0.21072397   0.7502911
34 Average 0.3543318 0.1506584  0.08750859   0.6674737
35 Average 0.4334945 0.1510094  0.16727492   0.7385365
36 Average 0.3896957 0.1401590  0.12435787   0.6609141
37 Average 0.5171465 0.1718475  0.19600911   0.8479718
38 Average 0.5564719 0.1915810  0.20836580   0.9315053
39 Average 0.5076128 0.1743942  0.18505179   0.8763346
40 Average 0.4016411 0.1498184  0.11966459   0.6961573
41 Average 0.4669331 0.1869691  0.10734355   0.8404699
42 Average 0.3267297 0.1586578  0.03709506   0.6696817
43 Average 0.6095316 0.1758992  0.28837768   0.9504233
44 Average 0.3186873 0.1474396  0.05649283   0.6143986
45 Average 0.2424751 0.1832698 -0.07740702   0.6248991
46 Average 0.2757444 0.1547151 -0.03328899   0.5676227
47 Average 0.3918988 0.1341843  0.13747243   0.6613308
48 Average 0.5559171 0.1881179  0.21888652   0.9306877
49 Average 0.6207464 0.1774968  0.28739352   0.9587773
50 Average 0.5116313 0.1767828  0.19799749   0.8773017
51 Average 0.3658950 0.1276997  0.13782309   0.6355595
```

```r
plot(asyn)
```

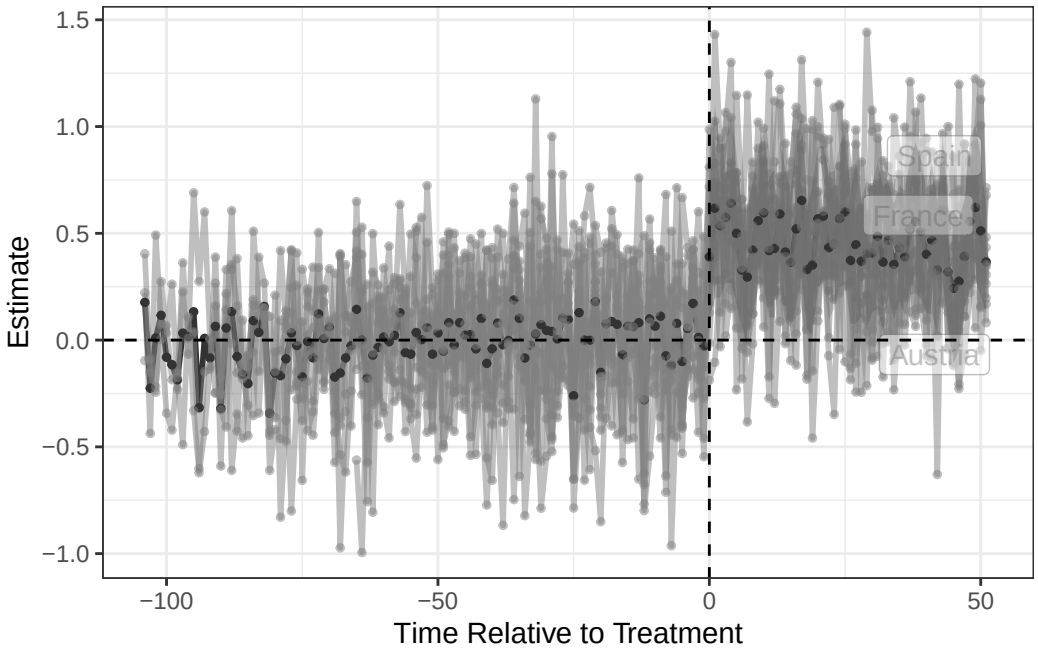
```
Joining with `by = join_by(Level)`

Warning: The `<scale>` argument of `guides()` cannot be `FALSE`. Use "none" instead as
of ggplot2 3.3.4.
i The deprecated feature was likely used in the augsynth package.
  Please report the issue to the authors.

Warning: Removed 272 rows containing missing values or values outside the scale range
(`geom_line()`).

Warning: Removed 272 rows containing missing values or values outside the scale range
(`geom_point()`).
```

```
Warning: ggrepel: 9 unlabeled data points (too many overlaps). Consider
increasing max.overlaps
```



Report:

- Overall **ATT**: 460, **SE**:0.132
- Since 79% improvement (global) > 55% improvement (individual), the algorithm was more successful at minimizing the pooled/global imbalance than the unit-specific imbalance.

6 6. Counterfactual Imputation Estimators

6.1 Fixed Effects counterfactual estimator (FEct)

```
library(fect)
```

```
Warning: package 'fect' was built under R version 4.5.1
```

```r
out_fe <- fect(
  watch_time_per_user ~ new_algo_deployed + marketing_spend + avg_monthly_price + broadband_p
  data   = df,
  index  = c("country","week"),
  method = "fe",
  force  = "two-way",
  se     = TRUE,
  nboots = 200,
  parallel = TRUE,
  cores    = 4
)
```

```
Parallel computing ...

Bootstrapping for uncertainties ...

200 runs
```

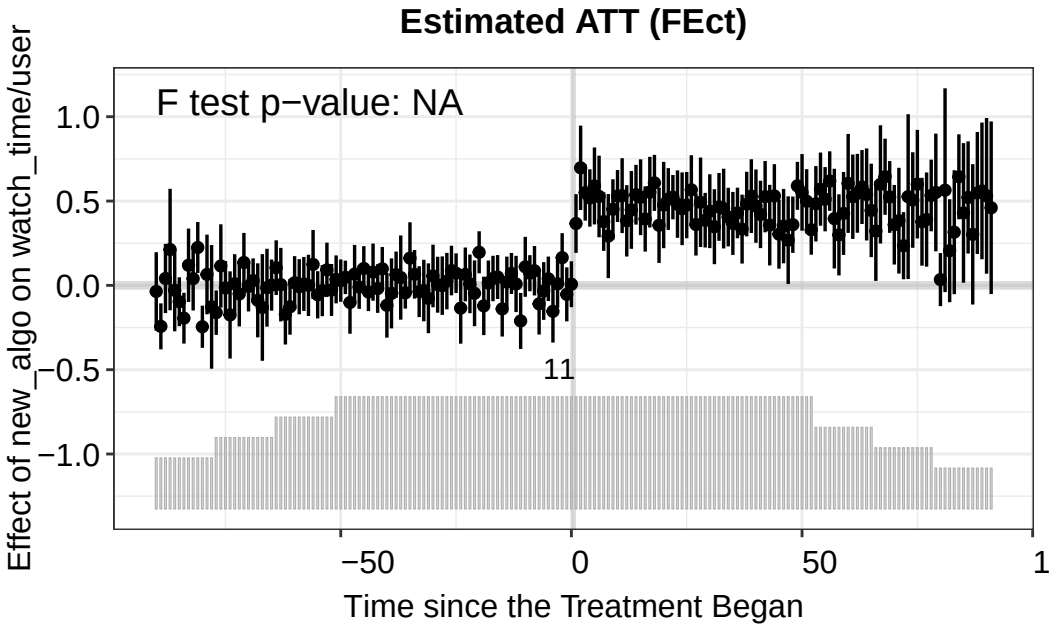
```r
summary(out_fe)
```

```
                     Length Class   Mode
Y.dat                 3120  -none-  numeric
D.dat                 3120  -none-  numeric
I.dat                 3120  -none-  numeric
Y                        1  -none-  character
D                        1  -none-  character
X                        3  -none-  character
T.on                  3120  -none-  numeric
G                        0  -none-  NULL
balance.period           0  -none-  NULL
hasRevs                  1  -none-  numeric
T.off                    0  -none-  NULL
T.on.balance          3120  -none-  logical
index                    2  -none-  character
id                      20  -none-  character
rawtime                156  Date    numeric
binary                   1  -none-  logical
loo                      1  -none-  logical
proportion               1  -none-  numeric
pre.periods             91  -none-  numeric
```

```
tost.threshold           1  -none-  numeric
placeboTest              1  -none-  logical
placebo.period           0  -none-  NULL
carryoverTest            1  -none-  logical
carryover.period         0  -none-  NULL
unit.type               20  -none-  numeric
obs.missing           3120  -none-  numeric
obs.missing.balance      1  -none-  logical
method                   1  -none-  character
Y.ct                  3120  -none-  numeric
Y.ct.full             3120  -none-  numeric
D                     3120  -none-  numeric
Y                     3120  -none-  numeric
X                     9360  -none-  numeric
eff                   3120  -none-  numeric
I                     3120  -none-  numeric
II                    3120  -none-  numeric
att.avg                  1  -none-  numeric
att.avg.unit             1  -none-  numeric
force                    1  -none-  numeric
T                        1  -none-  numeric
N                        1  -none-  numeric
Ntr                      1  -none-  numeric
Nco                      1  -none-  numeric
tr                      11  -none-  numeric
co                       9  -none-  numeric
p                        1  -none-  numeric
r.cv                     1  -none-  numeric
IC                       1  -none-  numeric
beta                     3  -none-  numeric
est                     11  -none-  list
mu                       1  -none-  numeric
niter                    1  -none-  numeric
validX                   1  -none-  numeric
validF                   1  -none-  numeric
time                   208  -none-  numeric
att                    208  -none-  numeric
count                  208  -none-  numeric
eff.calendar           312  -none-  numeric
N.calendar             156  -none-  numeric
eff.calendar.fit       312  -none-  numeric
calendar.enp             1  -none-  numeric
eff.pre               2652  -none-  numeric
```

```
eff.pre.equiv         2652  -none-  numeric
pre.sd                 312  -none-  numeric
PC                       1  -none-  numeric
sigma2                   1  -none-  numeric
sigma2.fect              1  -none-  numeric
res                   3120  -none-  numeric
res.full              3120  -none-  numeric
rmse                     1  -none-  numeric
alpha                   20  -none-  numeric
xi                     156  -none-  numeric
alpha.tr                11  -none-  numeric
alpha.co                 9  -none-  numeric
est.avg                  5  -none-  numeric
att.avg.boot           200  -none-  numeric
est.avg.unit             5  -none-  numeric
att.avg.unit.boot      200  -none-  numeric
est.eff.calendar       936  -none-  numeric
est.eff.calendar.fit   936  -none-  numeric
est.att               1248  -none-  numeric
att.bound              416  -none-  numeric
att.boot             41600  -none-  numeric
att.count.boot       41600  -none-  numeric
beta.boot              600  -none-  numeric
est.beta                15  -none-  numeric
test.out                 9  -none-  list
call                    10  -none-  call
formula                  3  formula call
```

```r
# plot results
plot(
  out_fe,
  main = "Estimated ATT (FEct)",
  ylab = "Effect of new_algo on watch_time/user",
  cex.main = 0.8, cex.lab = 0.8, cex.axis = 0.8,
  stats = "F.p"
)
```



6.2 Interactive Fixed Effects counterfactual estimator (iFEct)

```
library(fect)

out_ife <- fect(
  watch_time_per_user ~ new_algo_deployed + marketing_spend + avg_monthly_price + broadband_p
  data   = df,
  index  = c("country","week"),
  method = "ife",                  # interactive fixed effects
  force  = "two-way",
  se     = TRUE,
  nboots = 200,
  parallel = TRUE,
  cores    = 4,
  CV       = TRUE,
  r        = c(0,5),
  criterion = "mspe"
)
```

```
Parallel computing ...
```

```
Cross-validating ...

Criterion: Mean Squared Prediction Error

Interactive fixed effects model...


 r = 0; sigma2 = 0.09119; IC = -1.79297; PC = 0.08410; MSPE = 0.10110

*


 r = 1; sigma2 = 0.08689; IC = -1.25640; PC = 0.13121; MSPE = 0.11689


 r = 2; sigma2 = 0.08401; IC = -0.71198; PC = 0.17633; MSPE = 0.13303


 r = 3; sigma2 = 0.08069; IC = -0.18091; PC = 0.21694; MSPE = 0.15198


 r = 4; sigma2 = 0.07696; IC = 0.33631; PC = 0.25235; MSPE = 0.16391


 r = 5; sigma2 = 0.07357; IC = 0.84911; PC = 0.28474; MSPE = 0.18807


 r* = 0



Bootstrapping for uncertainties ...

200 runs
```

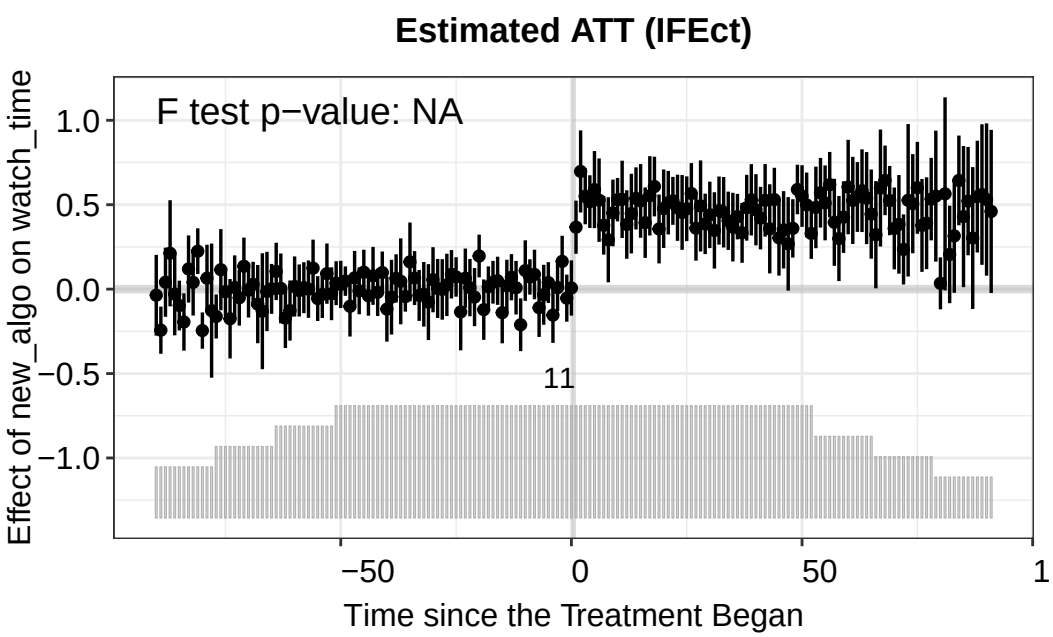
```r
summary(out_ife)
```

```
                      Length Class   Mode
Y.dat                  3120  -none-  numeric
D.dat                  3120  -none-  numeric
I.dat                  3120  -none-  numeric
Y                         1  -none-  character
D                         1  -none-  character
X                         3  -none-  character
T.on                   3120  -none-  numeric
G                         0  -none-  NULL
balance.period            0  -none-  NULL
hasRevs                   1  -none-  numeric
T.off                     0  -none-  NULL
T.on.balance           3120  -none-  logical
index                     2  -none-  character
id                       20  -none-  character
rawtime                 156  Date    numeric
binary                    1  -none-  logical
loo                       1  -none-  logical
proportion                1  -none-  numeric
pre.periods              91  -none-  numeric
tost.threshold            1  -none-  numeric
placeboTest               1  -none-  logical
placebo.period            0  -none-  NULL
carryoverTest             1  -none-  logical
carryover.period          0  -none-  NULL
unit.type                20  -none-  numeric
obs.missing            3120  -none-  numeric
obs.missing.balance       1  -none-  logical
sigma2                    1  -none-  numeric
sigma2.fect               1  -none-  numeric
T.on                   3120  -none-  numeric
Y.ct                   3120  -none-  numeric
eff                    3120  -none-  numeric
att.avg                   1  -none-  numeric
att.avg.unit              1  -none-  numeric
force                     1  -none-  numeric
T                         1  -none-  numeric
N                         1  -none-  numeric
Ntr                       1  -none-  numeric
Nco                       1  -none-  numeric
```

```
p                       1  -none-  numeric
D                    3120  -none-  numeric
Y                    3120  -none-  numeric
X                    9360  -none-  numeric
I                    3120  -none-  numeric
II                   3120  -none-  numeric
tr                     11  -none-  numeric
co                      9  -none-  numeric
est                    11  -none-  list
method                  1  -none-  character
mu                      1  -none-  numeric
beta                    3  -none-  numeric
validX                  1  -none-  numeric
validF                  1  -none-  numeric
niter                   1  -none-  numeric
time                  208  -none-  numeric
att                   208  -none-  numeric
count                 208  -none-  numeric
eff.calendar          312  -none-  numeric
N.calendar            156  -none-  numeric
eff.calendar.fit      312  -none-  numeric
calendar.enp            1  -none-  numeric
eff.pre              2652  -none-  numeric
eff.pre.equiv        2652  -none-  numeric
pre.sd                312  -none-  numeric
rmse                    1  -none-  numeric
rmCV                   10  -none-  list
estCV                  10  -none-  list
res                  3120  -none-  numeric
alpha                  20  -none-  numeric
xi                    156  -none-  numeric
alpha.tr               11  -none-  numeric
alpha.co                9  -none-  numeric
r.cv                    1  -none-  numeric
IC                      1  -none-  numeric
PC                      1  -none-  numeric
CV.out.ife             78  -none-  numeric
est.avg                 5  -none-  numeric
att.avg.boot          200  -none-  numeric
est.avg.unit            5  -none-  numeric
att.avg.unit.boot     200  -none-  numeric
est.eff.calendar      936  -none-  numeric
est.eff.calendar.fit  936  -none-  numeric
```

```
est.att                1248  -none-  numeric
att.bound               416  -none-  numeric
att.boot              41600  -none-  numeric
att.count.boot        41600  -none-  numeric
beta.boot               600  -none-  numeric
est.beta                 15  -none-  numeric
test.out                  9  -none-  list
call                     13  -none-  call
formula                   3  formula call
```

```r
# plot results
plot(
  out_ife,
  main = "Estimated ATT (IFEct)",
  ylab = "Effect of new_algo on watch_time",
  cex.main = 0.8, cex.lab = 0.8, cex.axis = 0.8,
  stats = "F.p"
)
```



- The cross-validation has suggested latent factors r=0(0.10453).

6.3 Matrix Completion counterfactual estimator (MC)

```r
library(fect)

out_mc <- fect(
  watch_time_per_user ~ new_algo_deployed + marketing_spend + avg_monthly_price + broadband_p
  data   = df,
  index  = c("country","week"),
  method = "mc",
  force  = "two-way",
  se     = TRUE,
  nboots = 200,
  parallel = TRUE,
  cores    = 4,
  CV       = TRUE,
  nlambda  = 10,
  criterion = "mspe"
)
```

```
Parallel computing ...

Cross-validating ...

Criterion: Mean Squared Prediction Error

Matrix completion method...


 lambda.norm = 1.00000; MSPE = 0.09385; MSPTATT = 0.00863; MSE = 0.09104

*


 lambda.norm = 0.42170; MSPE = 0.09529; MSPTATT = 0.00394; MSE = 0.04029


 lambda.norm = 0.17783; MSPE = 0.09451; MSPTATT = 0.00075; MSE = 0.00740
```

```
 lambda.norm = 0.07499; MSPE = 0.09380; MSPTATT = 0.00013; MSE = 0.00131



 lambda.norm* = 1



Bootstrapping for uncertainties ...

200 runs
```

```r
# report
summary(out_mc)
```

```
                     Length Class   Mode
Y.dat                 3120  -none-  numeric
D.dat                 3120  -none-  numeric
I.dat                 3120  -none-  numeric
Y                        1  -none-  character
D                        1  -none-  character
X                        3  -none-  character
T.on                  3120  -none-  numeric
G                        0  -none-  NULL
balance.period           0  -none-  NULL
hasRevs                  1  -none-  numeric
T.off                    0  -none-  NULL
T.on.balance          3120  -none-  logical
index                    2  -none-  character
id                      20  -none-  character
rawtime                156  Date    numeric
binary                   1  -none-  logical
loo                      1  -none-  logical
proportion               1  -none-  numeric
pre.periods             91  -none-  numeric
tost.threshold           1  -none-  numeric
placeboTest              1  -none-  logical
placebo.period           0  -none-  NULL
carryoverTest            1  -none-  logical
```

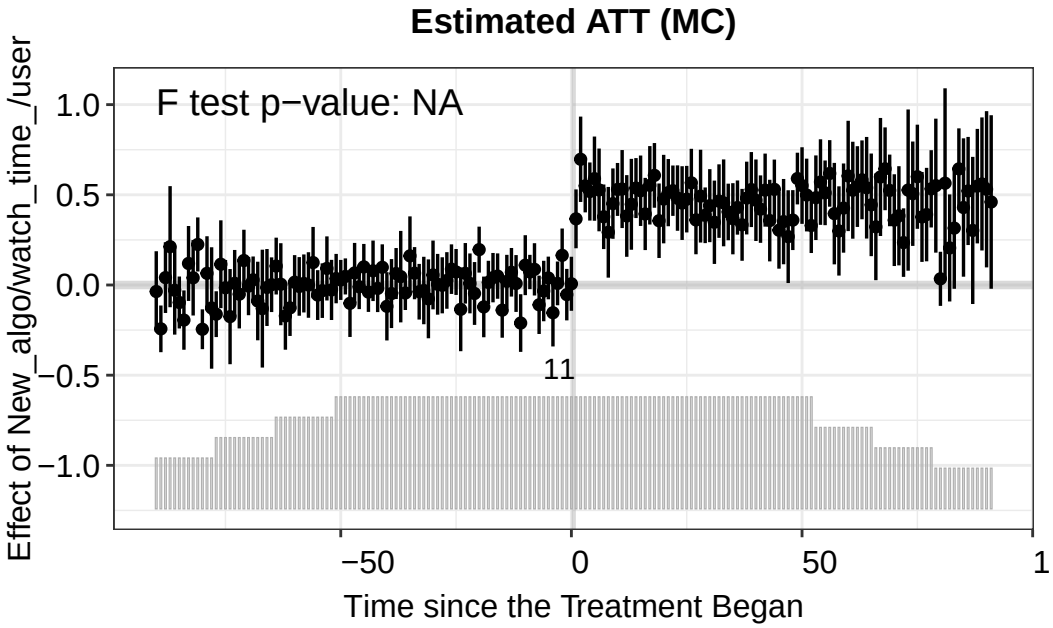
```
carryover.period         0  -none-  NULL
unit.type               20  -none-  numeric
obs.missing           3120  -none-  numeric
obs.missing.balance      1  -none-  logical
sigma2                   1  -none-  numeric
sigma2.fect              1  -none-  numeric
T.on                  3120  -none-  numeric
Y.ct                  3120  -none-  numeric
eff                   3120  -none-  numeric
att.avg                  1  -none-  numeric
att.avg.unit             1  -none-  numeric
force                    1  -none-  numeric
T                        1  -none-  numeric
N                        1  -none-  numeric
Ntr                      1  -none-  numeric
Nco                      1  -none-  numeric
p                        1  -none-  numeric
D                     3120  -none-  numeric
Y                     3120  -none-  numeric
X                     9360  -none-  numeric
I                     3120  -none-  numeric
II                    3120  -none-  numeric
tr                      11  -none-  numeric
co                       9  -none-  numeric
est                     10  -none-  list
method                   1  -none-  character
mu                       1  -none-  numeric
beta                     3  -none-  numeric
validX                   1  -none-  numeric
validF                   1  -none-  numeric
niter                    1  -none-  numeric
time                   208  -none-  numeric
att                    208  -none-  numeric
count                  208  -none-  numeric
eff.calendar           312  -none-  numeric
N.calendar             156  -none-  numeric
eff.calendar.fit       312  -none-  numeric
calendar.enp             1  -none-  numeric
eff.pre               2652  -none-  numeric
eff.pre.equiv         2652  -none-  numeric
pre.sd                 312  -none-  numeric
rmse                     1  -none-  numeric
rmCV                    10  -none-  list
```

```
estCV                  10  -none-  list
res                  3120  -none-  numeric
alpha                  20  -none-  numeric
xi                    156  -none-  numeric
alpha.tr               11  -none-  numeric
alpha.co                9  -none-  numeric
lambda.cv               1  -none-  numeric
lambda.seq             10  -none-  numeric
lambda.norm             1  -none-  numeric
eigen.all              20  -none-  numeric
CV.out.mc             100  -none-  numeric
est.avg                 5  -none-  numeric
att.avg.boot          200  -none-  numeric
est.avg.unit            5  -none-  numeric
att.avg.unit.boot     200  -none-  numeric
est.eff.calendar      936  -none-  numeric
est.eff.calendar.fit  936  -none-  numeric
est.att              1248  -none-  numeric
att.bound             416  -none-  numeric
att.boot            41600  -none-  numeric
att.count.boot      41600  -none-  numeric
beta.boot             600  -none-  numeric
est.beta               15  -none-  numeric
test.out                9  -none-  list
call                   13  -none-  call
formula                 3  formula call
```

```r
# chosen penalty (lambda*)
out_mc$lambda.cv
```

```
[1] 0.001433092
```

```r
# plot results
plot(
  out_mc,
  main = "Estimated ATT (MC)",
  ylab = "Effect of New_algo/watch_time_/user",
  cex.main = 0.8, cex.lab = 0.8, cex.axis = 0.8,
  stats = "F.p"
)
```



- Cross-validation suggested a nuclear-norm penalty of = 1 (lambda.norm = 1.00000).

Best Fit analysis:

The Matrix Completion (MC) model is best because it has the smallest error (MSPE = 0.10208). In cross-validation, = 1 gave the lowest prediction error, as a result, MC's counterfactual predictions matched the pre-treatment outcomes more closely than FEct or iFEct.

7 7. Parallel Trends Assumption

7.1 Preferred model section:

```
library(fect)

out <- fect(
  watch_time_per_user ~ new_algo_deployed + marketing_spend + avg_monthly_price + broadband_p
  data    = df,
  index   = c("country","week"),
  method = "ife",
  se = TRUE
)
```

```
Parallel computing ...

Bootstrapping for uncertainties ...

200 runs
```

```r
summary(out)
```

```
                     Length Class   Mode
Y.dat                 3120  -none-  numeric
D.dat                 3120  -none-  numeric
I.dat                 3120  -none-  numeric
Y                        1  -none-  character
D                        1  -none-  character
X                        3  -none-  character
T.on                  3120  -none-  numeric
G                        0  -none-  NULL
balance.period           0  -none-  NULL
hasRevs                  1  -none-  numeric
T.off                    0  -none-  NULL
T.on.balance          3120  -none-  logical
index                    2  -none-  character
id                      20  -none-  character
rawtime                156  Date    numeric
binary                   1  -none-  logical
loo                      1  -none-  logical
proportion               1  -none-  numeric
pre.periods             91  -none-  numeric
tost.threshold           1  -none-  numeric
placeboTest              1  -none-  logical
placebo.period           0  -none-  NULL
carryoverTest            1  -none-  logical
carryover.period         0  -none-  NULL
unit.type               20  -none-  numeric
obs.missing           3120  -none-  numeric
obs.missing.balance      1  -none-  logical
method                   1  -none-  character
Y.ct                  3120  -none-  numeric
Y.ct.full             3120  -none-  numeric
D                     3120  -none-  numeric
Y                     3120  -none-  numeric
```

```
X                    9360  -none-  numeric
eff                  3120  -none-  numeric
I                    3120  -none-  numeric
II                   3120  -none-  numeric
att.avg                 1  -none-  numeric
att.avg.unit            1  -none-  numeric
force                   1  -none-  numeric
T                       1  -none-  numeric
N                       1  -none-  numeric
Ntr                     1  -none-  numeric
Nco                     1  -none-  numeric
tr                     11  -none-  numeric
co                      9  -none-  numeric
p                       1  -none-  numeric
r.cv                    1  -none-  numeric
IC                      1  -none-  numeric
beta                    3  -none-  numeric
est                    11  -none-  list
mu                      1  -none-  numeric
niter                   1  -none-  numeric
validX                  1  -none-  numeric
validF                  1  -none-  numeric
time                  208  -none-  numeric
att                   208  -none-  numeric
count                 208  -none-  numeric
eff.calendar          312  -none-  numeric
N.calendar            156  -none-  numeric
eff.calendar.fit      312  -none-  numeric
calendar.enp            1  -none-  numeric
eff.pre              2652  -none-  numeric
eff.pre.equiv        2652  -none-  numeric
pre.sd                312  -none-  numeric
PC                      1  -none-  numeric
sigma2                  1  -none-  numeric
sigma2.fect             1  -none-  numeric
res                  3120  -none-  numeric
res.full             3120  -none-  numeric
rmse                    1  -none-  numeric
alpha                  20  -none-  numeric
xi                    156  -none-  numeric
alpha.tr               11  -none-  numeric
alpha.co                9  -none-  numeric
est.avg                 5  -none-  numeric
```

```
att.avg.boot           200  -none-  numeric
est.avg.unit             5  -none-  numeric
att.avg.unit.boot      200  -none-  numeric
est.eff.calendar       936  -none-  numeric
est.eff.calendar.fit   936  -none-  numeric
est.att               1248  -none-  numeric
att.bound              416  -none-  numeric
att.boot             41600  -none-  numeric
att.count.boot       41600  -none-  numeric
beta.boot              600  -none-  numeric
est.beta                15  -none-  numeric
test.out                 9  -none-  list
call                     6  -none-  call
formula                  3  formula call
```

7.2 Placebo test

```r
library(fect)
out_placebo <- fect(
  watch_time_per_user ~ new_algo_deployed + marketing_spend + avg_monthly_price + broadband_p
  data   = df,
  index  = c("country","week"),
  method = "ife",
  se     = TRUE,
  placeboTest   = TRUE,
  placebo.period = -10:-1
  # pick available pre-treatment periods
)
```

```
Error in fect.default(formula = NULL, data = data, Y = Yname, D = Dname, : "placebo.period" c
```

```r
summary(out_placebo)
```

```
Error: object 'out_placebo' not found
```

```r
plot(out_placebo, type = "placebo")
```

```
Error: object 'out_placebo' not found
```

7.3 Sensitivity analysis

```r
devtools::install_github("xuyiqing/fect")
```

```
WARNING: Rtools is required to build R packages, but no version of Rtools compatible with R 4

Please download and install Rtools 4.5 from https://cran.r-project.org/bin/windows/Rtools/.

Using GitHub PAT from the git credential store.

Downloading GitHub repo xuyiqing/fect@HEAD

WARNING: Rtools is required to build R packages, but is not currently installed.

Please download and install the appropriate version of Rtools for 4.5.0 from
https://cran.r-project.org/bin/windows/Rtools/.

Error: Failed to install 'fect' from GitHub:
  Could not find tools necessary to compile a package
Call `pkgbuild::check_build_tools(debug = TRUE)` to diagnose the problem.
```

```r
library(fect)
out_ife_placebo <- fect_sens(
  fect.out      = out,                # model with placebo
  post.periods  = 1:10,                     # focus on post-treatment periods
  Mbarvec       = seq(0, 0.5, by = 0.05),  # Relative Magnitude restriction grid
  periodMbarvec = c(0.15, 0.25),           # Period-specific restrictions
  Mvec          = seq(0, 0.2, by = 0.02),  # Smoothness restriction grid
  periodMvec    = c(0, 0.1),               # Period-specific smoothness
  parallel      = FALSE
)
```

```
Error in fect_sens(fect.out = out, post.periods = 1:10, Mbarvec = seq(0, : could not find fu
```

```r
#plotting
plot(out_ife_placebo,
     type = "sens",
     restrict = "rm",   # Relative Magnitude restriction
     main = "Relative Magnitude Restriction for ATT")
```

```
Error: object 'out_ife_placebo' not found
```